

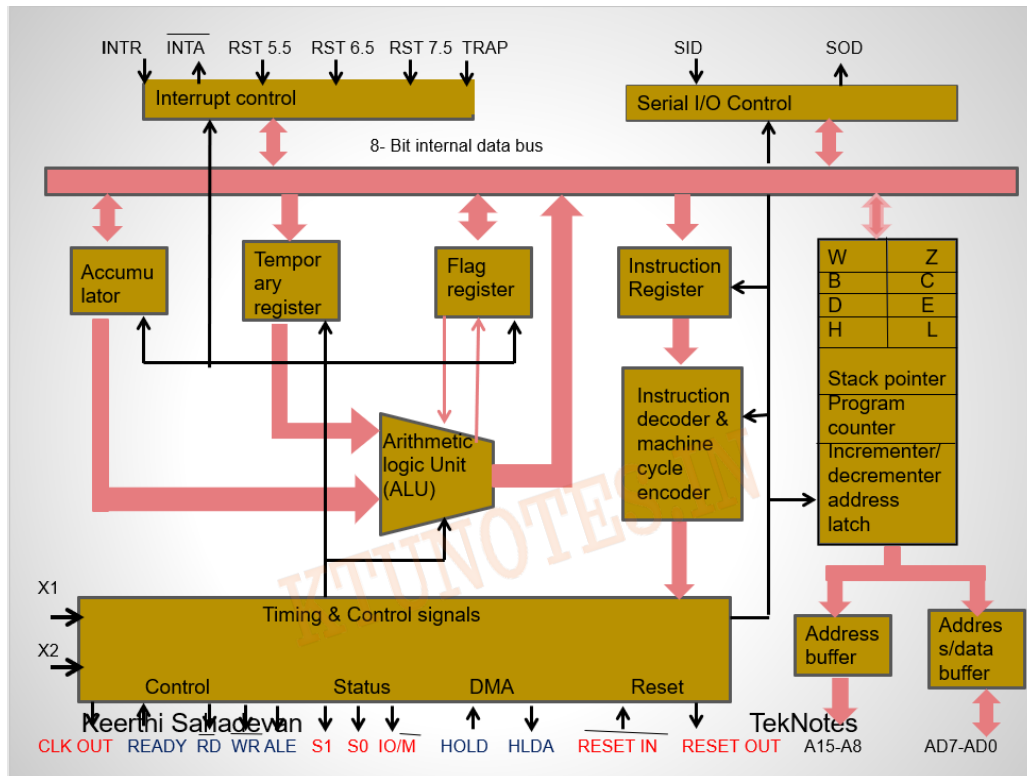
## Previous University questions & Answers

### Module-1

December 2017

1. Explain the architecture of 8085 microprocessor with neat block diagram (10)

**Answer:**



Microprocessor is a programmable integrated device that has computing and decision making capability. 8085 is 8-bit microprocessor. It mainly contains ALU, timing and control unit and registers. It has 8 bit data bus and 16 bit address bus.

**Registers:** Used to store data temporarily during the execution of program

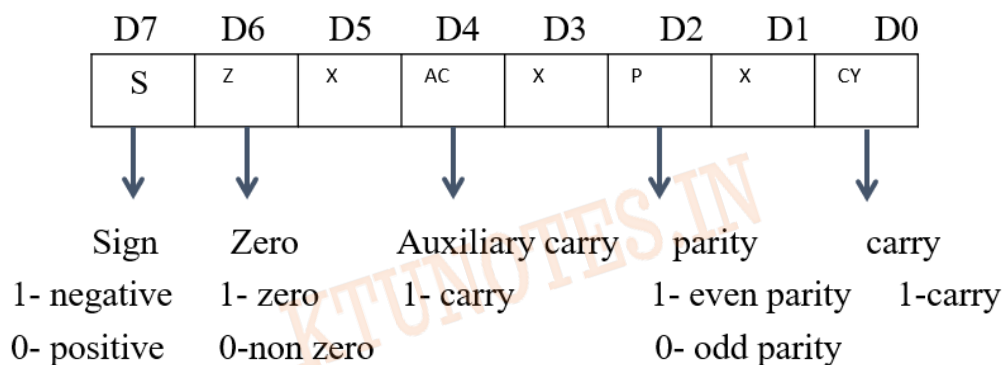
|                      |               |
|----------------------|---------------|
| W Register           | Z Register    |
| A Register           | Flag Register |
| B Register           | C Register    |
| D Register           | E Register    |
| H Register           | L Register    |
| Stack Pointer (SP)   |               |
| Program Counter (PC) |               |

**General purpose registers:** The 8085 has six general purpose registers to store 8-bit data. They are denoted as B, C, D, E, H and L. They can be combined as register pair – BC, DE, HL – to perform 16-bit operations.

**Special purpose register:**

**1.Accumulator (A):** The accumulator is an 8-bit register used to store 8-bit data and to perform arithmetic and logical instructions. Accumulator is also used for input/output operations.

**2.Flags:**



There are five flip-flops which can be set or reset after an operation according to data conditions of the result in accumulator and other registers. They are sign, zero, auxiliary carry, parity and carry.

**3.Instruction register:** It is used to fetch instructions.

**4. Program Counter:** It is used to store 16-bit address of next instruction to be fetched.

**5. Stack pointer:** 16-bit register used to store address of most recent stack entry.

**Temporary registers:** Programmer cannot access these registers. They are

- a) W & Z registers: Used to store data temporarily.
- b) Temporary data register: Internally used for most arithmetic & logical instructions.

**Incrementer /decrementer address latch:** 16- bit register used to increment or decrement the content of program counter or stack pointer

**Address buffer:** 8-bit **unidirectional** buffer used to drive higher order address bus (A15-A8)

**Address/data buffer:** 8-bit **bidirectional** buffer used to drive lower order address bus (A7-A0) & address bus (D7-D0)

**Interrupt control:** It has five interrupt inputs RST 7.5, RST 6.5, RST 5.5, INTR , TRAP and one interrupt acknowledge signal INTA.

**Serial input/output control:** In serial data communication one bit is transferred at a time over a single line. There are two pins to receive and transmit data. They are SOD and SID. Serial Output Data (SOD) line is used to send data serially. Serial Input Data (SID) line is used to receive data serially.

Timing and control unit: This unit is responsible for all operations in 8085. The functions are:

- Controls fetching & decoding of instructions.
- Generate appropriate signals for instruction execution.
- Generate signals required to interface external device to the processor.

The control signals are:

- Status signals: S0, S1,  $\overline{IO/\overline{M}}$
- DMA signals: HOLD, HLDA
- Reset signals:  $\overline{\text{RESET IN}}$ , RESET OUT
- Control signal:  $\overline{\text{RD}}$ ,  $\overline{\text{WR}}$ , ALE
- Clock: CLOCK OUT

## 2. Explain different addressing modes of 8085 with example. (6)

**Answer:** The different ways that a microprocessor can access data are referred to as addressing modes. 8085 has 5 addressing modes

- Immediate addressing mode: 8 or 16 bit data specified as a part of instruction. I indicate immediate addressing mode.

Examples:

MVI A, 34H

LXI D, 10FFH

ADI 45H

2. Register addressing mode: Specifies source operand, destination operand or both in 8085 registers.

Example: MOV A, B

ADD C

3. Direct addressing mode: 16-bit address of the operand within the instruction itself.

Examples: LDA 3000H

IN 07H

4. Indirect addressing mode: The memory address where the operand located is specified by the content of register pair.

Examples: LDAX B

MOV M, A

5. Implied addressing mode: Opcode specifies the address of operand

Examples: CMA

RAL

## December 2018

3. Explain the following pins in microprocessor

i) HOLD    ii) ready    iii) TRAP    iv) ALE    (5)

**Answer:**

**ALE:** Address Latch Enable. It goes high when operation is started by the processor. ALE is high, when address passed through address/data bus.

**READY:** Used by microprocessor to sense whether peripherals are ready or not for data transfer. It is used to synchronize slower external device to microprocessor. If it is low, then microprocessor waits until becomes high.

**HOLD:** It indicates that another device is requesting for the use of address bus, data bus & control bus. When microprocessor receives hold signal it completes current machine cycle & stops executing next instruction

**TRAP:** TRAP is a non-maskable interrupt. Whenever this pin gets activated, the 8085 always get interrupted. The input of TRAP is level sensitive and edge sensitive. It is highest priority interrupt.

4. Write an ALP in 8085 to find the smallest number in an array (5)

(Write the program of ascending order of an array)

```
REPEAT    MVI D, FFH
          LXI H, 3040H
          MOV B, M
          DCR B
          INX H
BACK      MOV A, M
          INX H
          CMP M
          JC NEXT BYTE
          MOV C, M
          MOV M, A
          DCX H
          MOV M, A
          INX H
          DCR D
NEXT BYTE DCR B
          JNZ BACK
          MOV A,D
          CPI FFH
          JNZ REPEAT
```

5. Explain with neat block diagram architecture of intel 8085 microprocessor. (10)

Answer: Repeated question

December 2019

6. Explain the operation of following instructions (5)

i) MVI C, 05H   ii) INR H   iii) MOV A, B   iv) CMA

**MVI C, 05H:** Load 8-bit data 05 to C register. It is an immediate addressing mode.

**INR H:** Increment the content of H register by one.

**MOV A, B:** Move the content of B register to A register

**CMA:** Complement the content of A register.

7. Explain the function of following pins in 8085

i) ALE   ii) TRAP   (4 marks)   - Repeated question

8. Write an ALP in 8085 to find largest number in an array (6)

Answer: Write the program in descending order of an array. Change the instruction JC by JNC in the program of ascending order.

```
REPEAT     MVI D, FFH  
            LXI H, 3040H  
            MOV B, M  
            DCR B  
            INX H  
BACK       MOV A, M  
            INX H  
            CMP M  
            JNC NEXT BYTE  
            MOV C, M  
            MOV M, A  
            DCX H  
            MOV M, A  
            INX H
```

DCR D

NEXT BYTE DCR B

JNZ BACK

MOV A, D

CPI FFH

JNZ REPEAT

**MAY 2019**

9. The content of accumulator and B register are 2AH and ABH respectively. Find the content of A register and flag after the execution of instruction ADD B. (5)

Answer:

$$\begin{array}{r} 2A + \\ AB \\ \hline D5 \end{array}$$
$$\begin{array}{r} 0010 \quad 1010 \quad + \\ 1010 \quad 1011 \\ \hline 1101 \quad 0101 \\ 13 \quad 5 \end{array}$$

Auxiliary carry - 1, carry - 0, parity - odd

flag content

| S | Z | AC | P | CY |
|---|---|----|---|----|
| 0 | 0 | 1  | 0 | 0  |

10. Explain the operation of following instruction in 8085: (5)

i) MOV A, M      ii) XCHG

Answer:

MOV A, M - Copy the data byte into register from the memory specified by the address in HL register pair

XCHG – Exchange the content of HL register pair with DE register pair. The content of H & D registers and L & E registers are swapped respectively.

**April 2018**

12. With examples explain the addressing modes of 8085. (5)-Repeated question

13. With a neat block diagram explain the internal architecture of 8085 microprocessor. (10)

Repeated question

14. Write a program to sort an array of 10 numbers in ascending order. (6) – Repeated question

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